Cover letter to the Editors of *Physical Review A*

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Dear Editors,

We are pleased to submit our manuscript, **"Catalytic quantum error correction: efficient catalyst preparation and a completely positive covariant recovery channel"**, for consideration as a Regular Article in *Physical Review A* (Quantum information).

Summary of the contribution. Shiraishi and Takagi (*Phys. Rev. Lett.* **132**, 180202, 2024) proved that catalytic covariant operations can amplify quantum coherence at an unbounded asymptotic rate. Turning that theorem into an operational state-recovery scheme has been blocked by two obstacles: the constructive proof requires $n^* \sim d^4 e^{2\gamma}$ noisy copies to prepare the catalyst (about 7.3×10^{10} for six qubits at dephasing $\gamma = 2$), and no concrete covariant recovery channel had been exhibited. This manuscript addresses both. (i) A three-stage pipeline—CPMG dynamical decoupling, Clifford twirling, recursive swap-test purification—prepares catalysts at the matched $F_{\mathrm{cat}} \geq 0.99$ endpoint from 32 copies, a 10^4–10^9-fold reduction depending on dimension, with postselection costs quantified ($P_{\mathrm{succ}} \approx 0.7$). (ii) An explicitly completely positive, exactly covariant joint-channel implementation of the recovery map (system–catalyst–ancilla; catalyst obtained by partial trace, never copied) achieves genuine fidelity recovery under dephasing at $d = 4$ ($0.54 \to 0.77$), passes a target-swap control experiment, and delimits the method's current reach: shallow covariant circuits do not yet help under depolarizing noise or at $d = 8$, and they consume rather than preserve the catalyst. An effective surrogate model of the asymptotic limit, benchmarked across 200 noise configurations ($d = 4$–64), is clearly separated from the channel-level results and collected in an appendix.

Why *Physical Review A*.

- *Protocol with explicit success conditions.* The CQEC protocol is formalized through the mode-inclusion criterion $\mathcal{C}(\rho_0) \subseteq \mathcal{C}(\rho_{\mathrm{noisy}})$, with finite-copy fidelity bounds and a clear statement of the operational assumptions (known target state, multiple copies), including its relation to stabilizer QEC, covariant codes, and the concurrent Purification QEC of Raghoonanan & Byrnes (arXiv:2603.11568).
- *Verified numerics.* Every quantitative property of the joint channel is locked in by unit tests: $\|[U, H_{\mathrm{total}}]\| = 0$ exactly, linearity to machine precision, Choi positivity, trace preservation, and catalyst-by-partial-trace.
- *Scoped claims.* Model-level and channel-level results are explicitly distinguished throughout; sensitivity of the pipeline's dimension independence to the twirl model is derived rather than hidden.
- *Open science.* The reference implementation is released as the `cqec` Python package v0.2.0 (MIT license) with a 52-test `pytest` suite and a one-command reproduction script: `github.com/deeptell-inc/cqec`.

Suggested referees (institutional emails from public faculty pages; please verify before contact):

- **Naoto Shiraishi** (The University of Tokyo; shiraishi@phys.c.u-tokyo.ac.jp) — co-author of the foundational PRL on catalytic coherence transformations.
- **Ryuji Takagi** (Nanyang Technological University; ryuji.takagi@ntu.edu.sg) — co-author of the same PRL; resource-theoretic content.
- **Tim Byrnes** (NYU Shanghai; tim.byrnes@nyu.edu) — co-author of the concurrent Purification QEC framework compared in our taxonomy table.
- **Iman Marvian** (Duke University; iman.marvian@duke.edu) — covariant operations and resource theory of asymmetry.
- **Andrew M. Childs** (University of Maryland; amchilds@umd.edu) — recursive swap-test purification adapted in our pipeline.

Statements. This work is not under consideration elsewhere. A complementary manuscript, "Blind Catalytic Quantum Error Correction" (independent content, no shared figures or theorems), is in review at a different journal. All authors have approved the submission; we declare no competing interests. The manuscript is archived at arXiv:2603.25774.

We thank you for your consideration and look forward to the referees' feedback.

Sincerely,

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